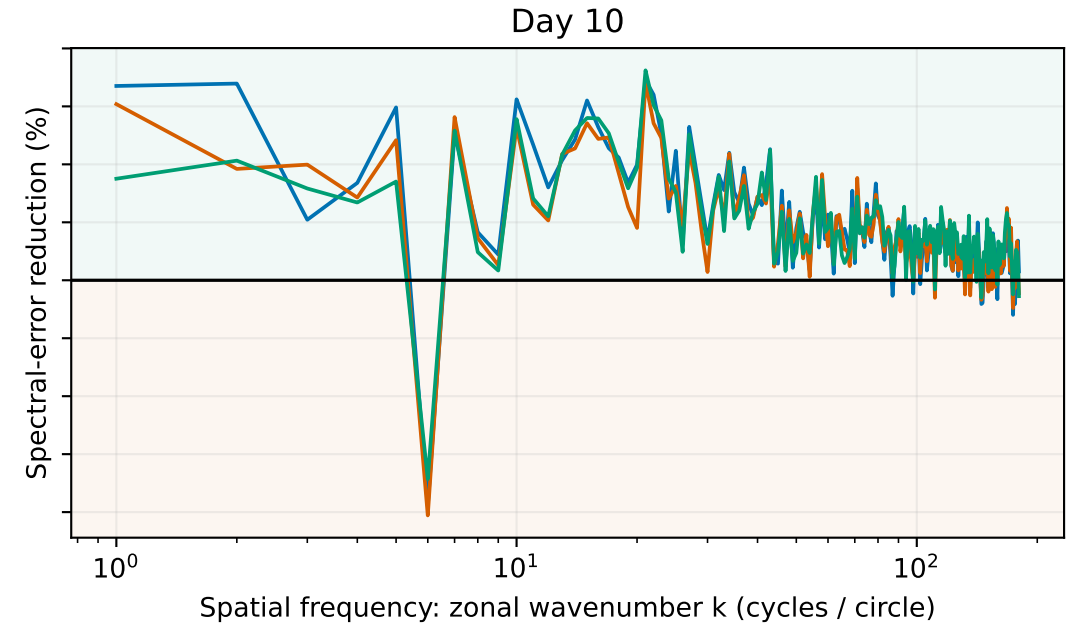
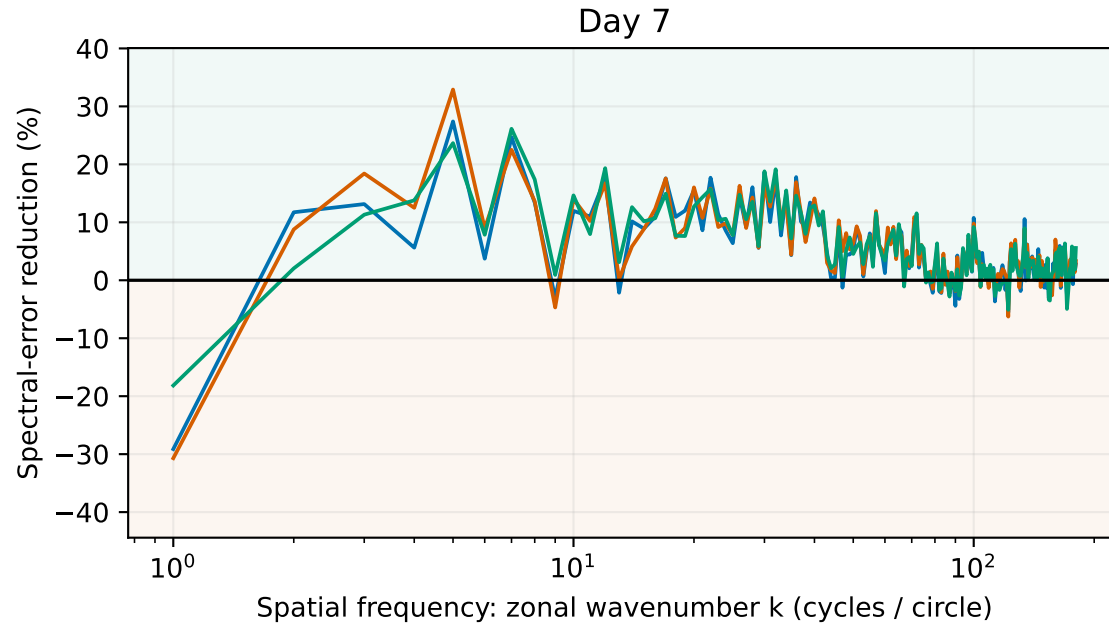
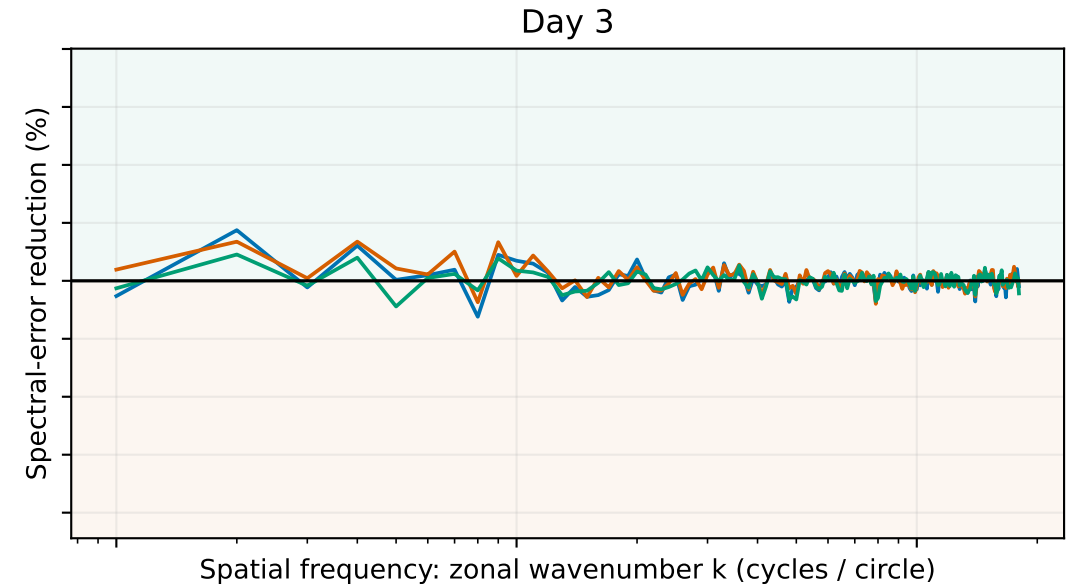
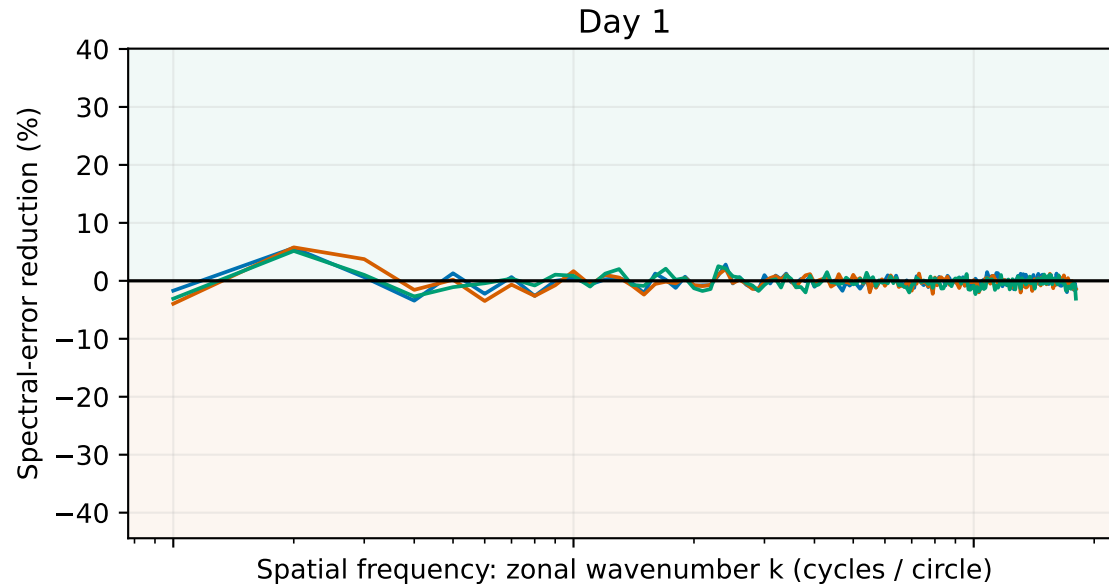


T2m: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



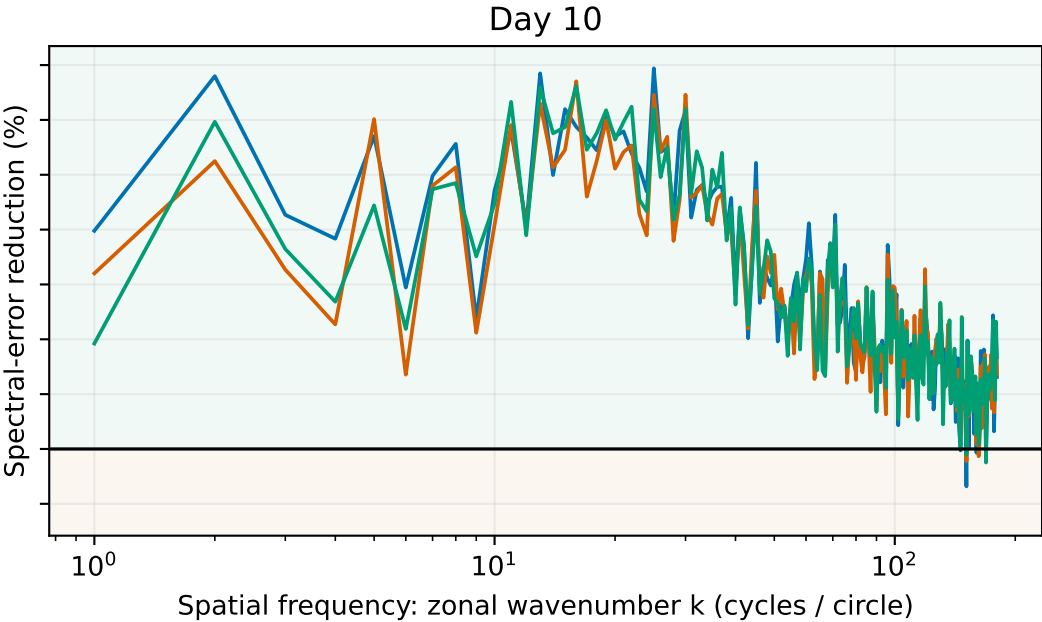
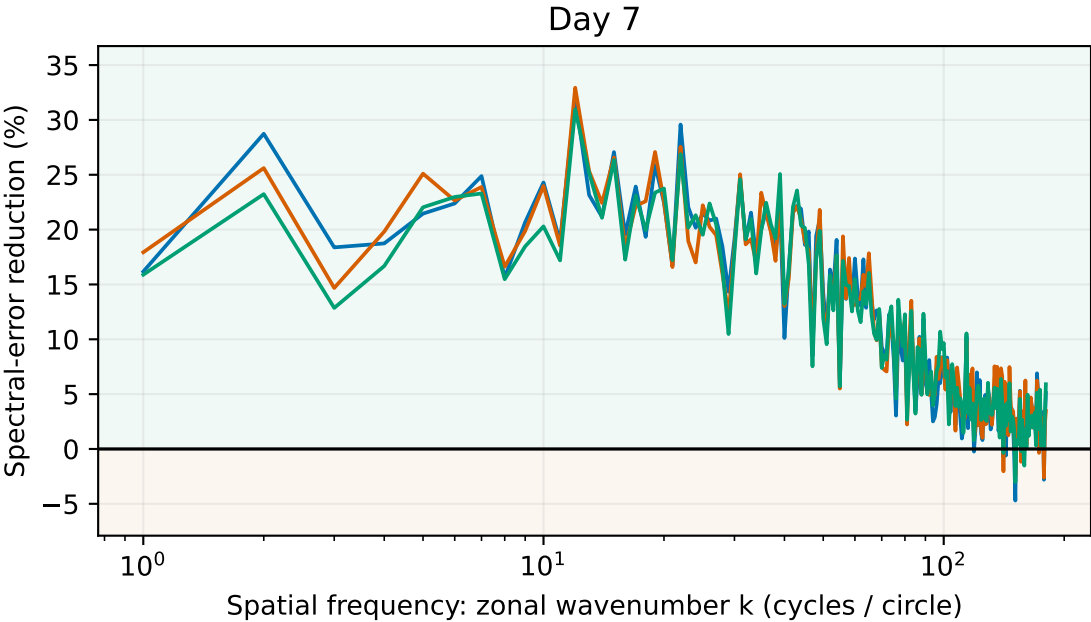
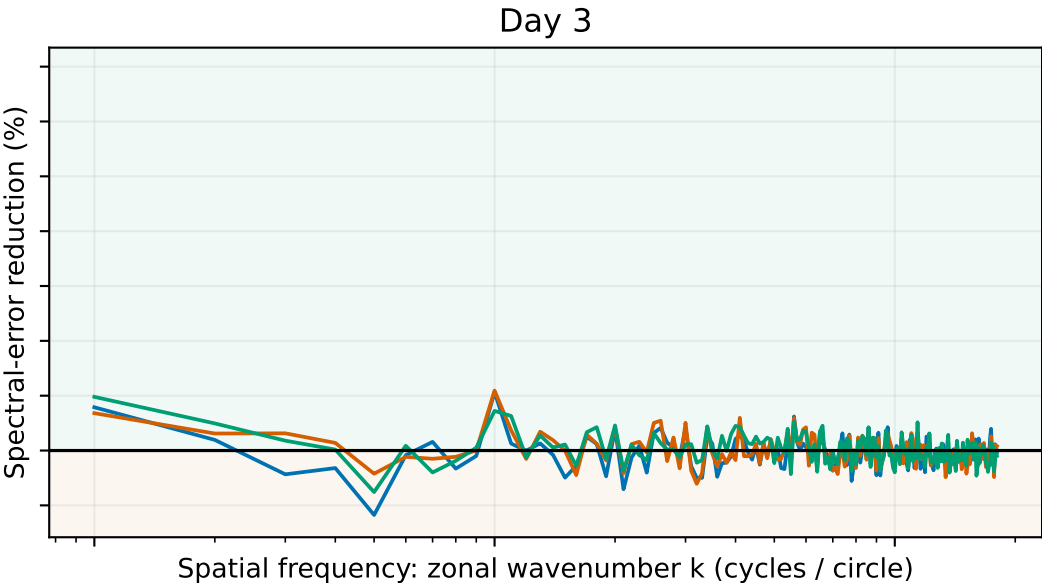
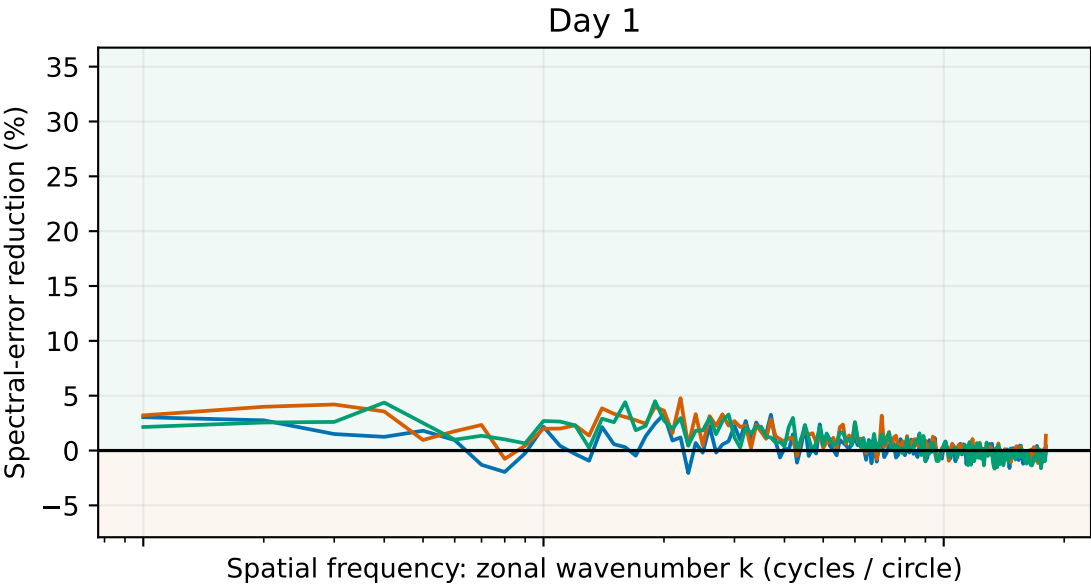
— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

U10: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



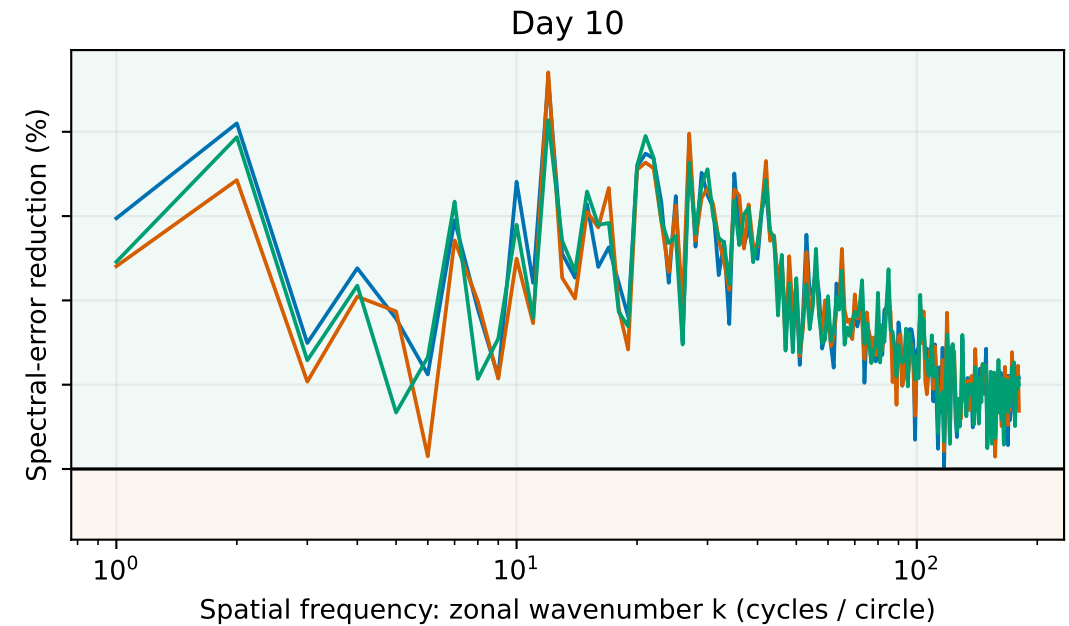
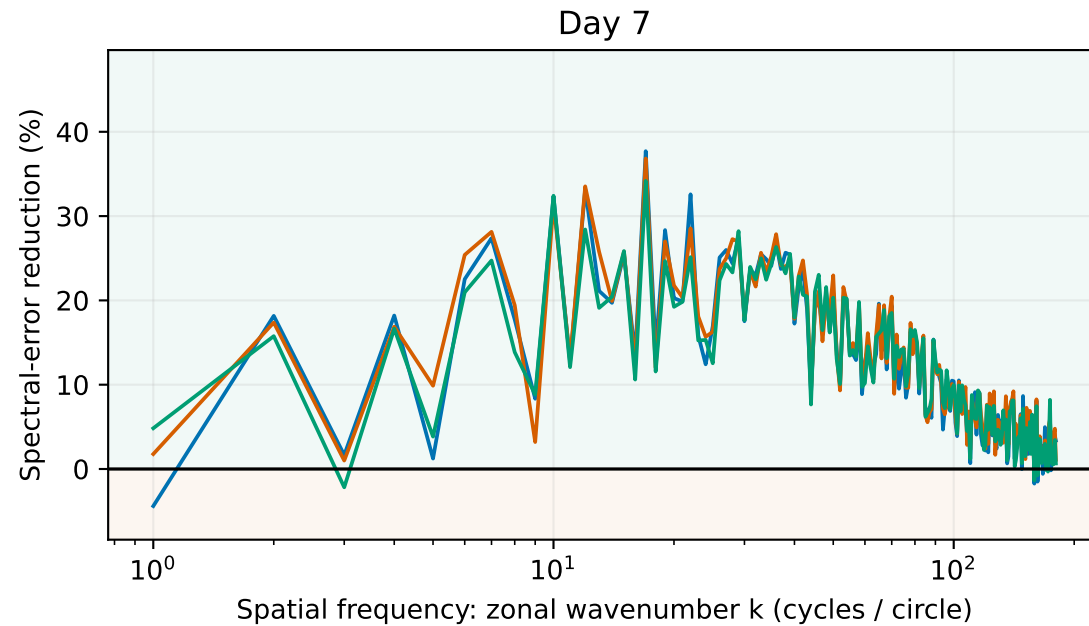
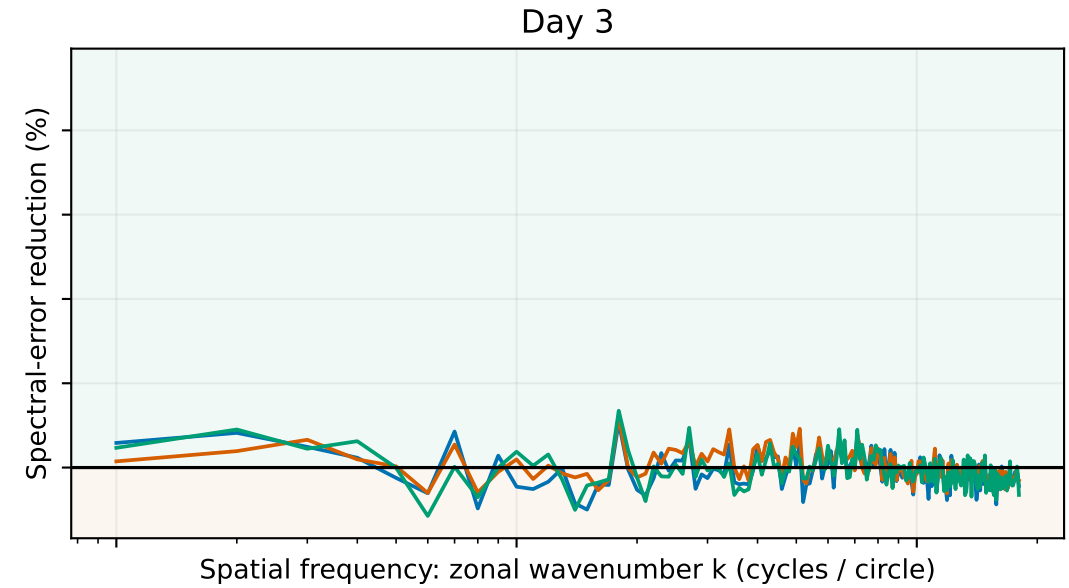
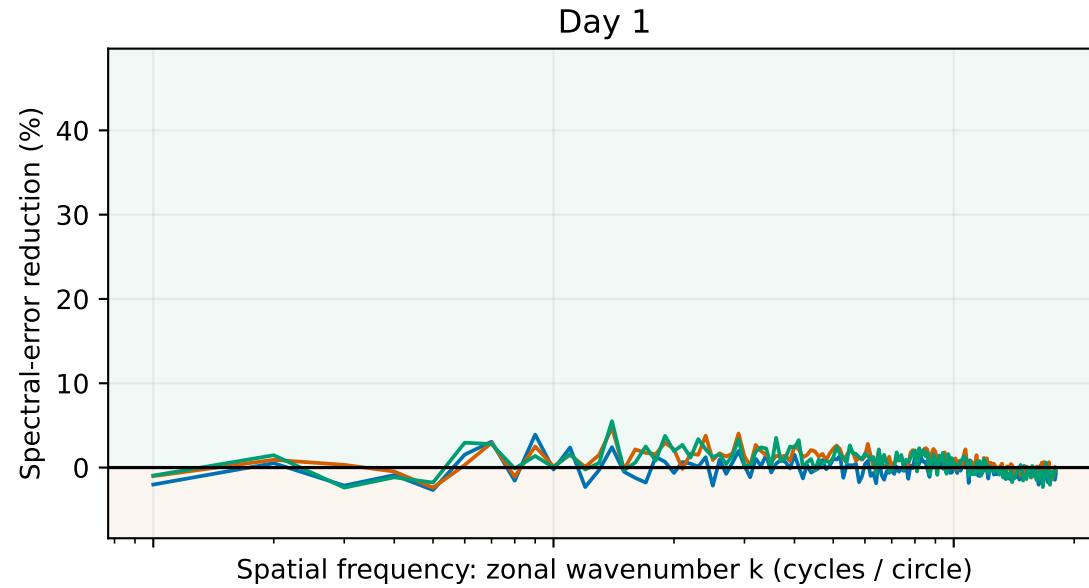
di16 · faster Mamba LR di32 · joint LR di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

V10: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



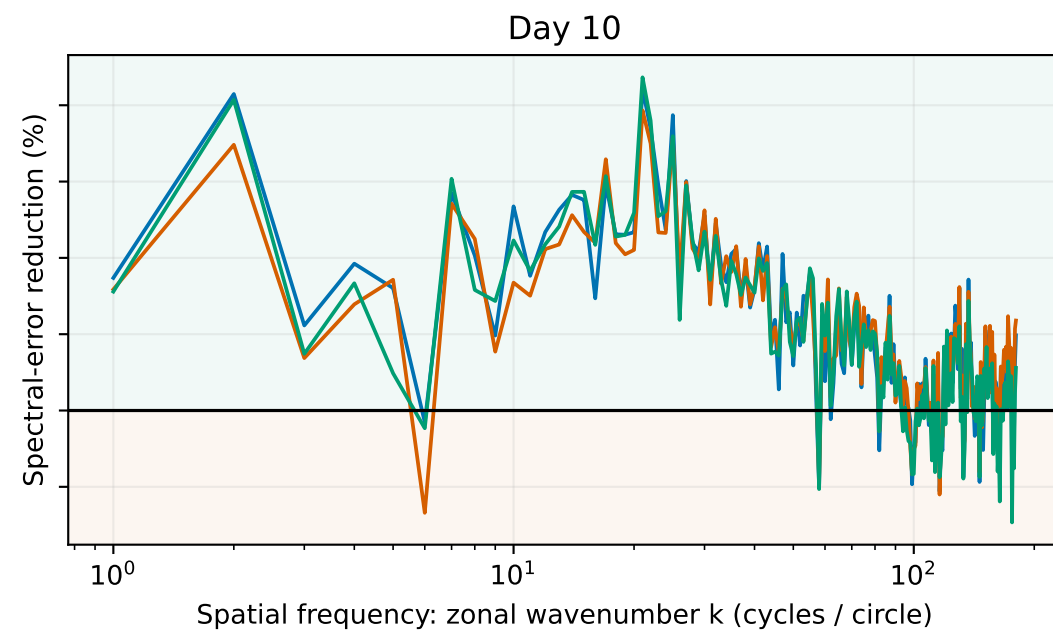
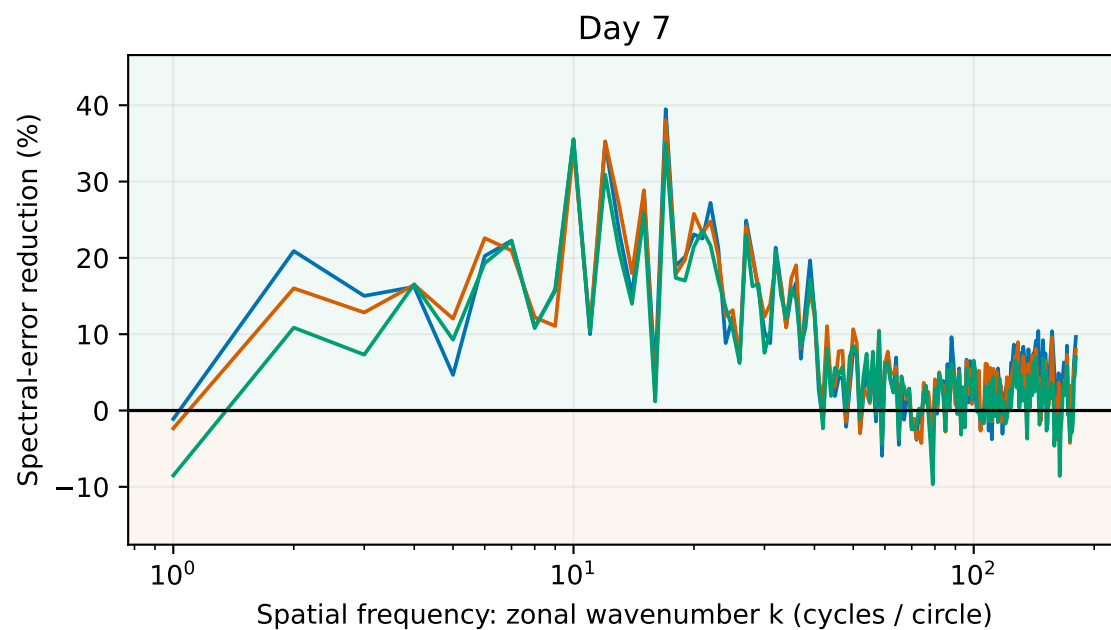
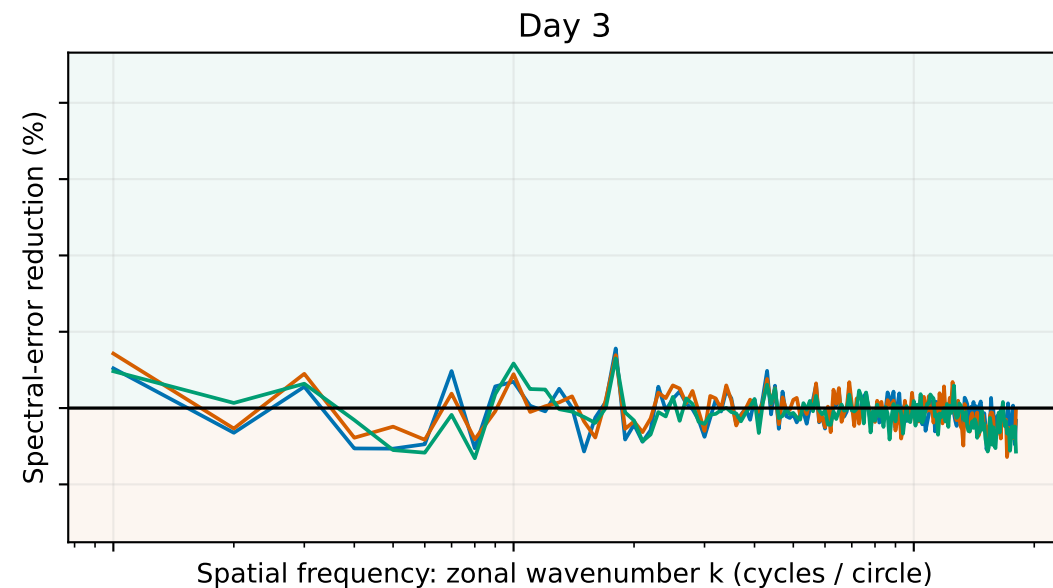
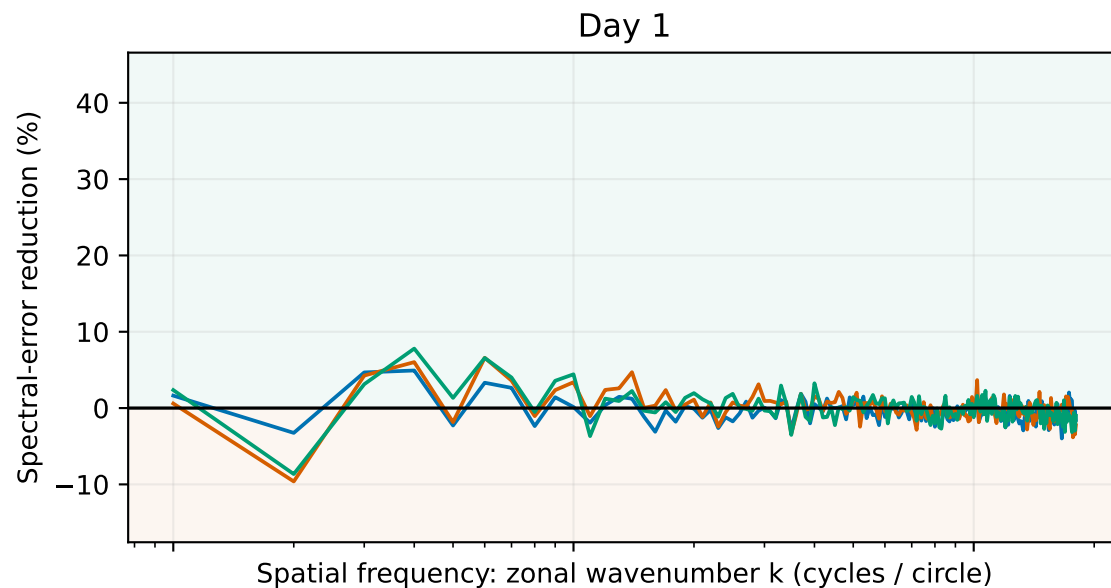
— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

MSLP: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



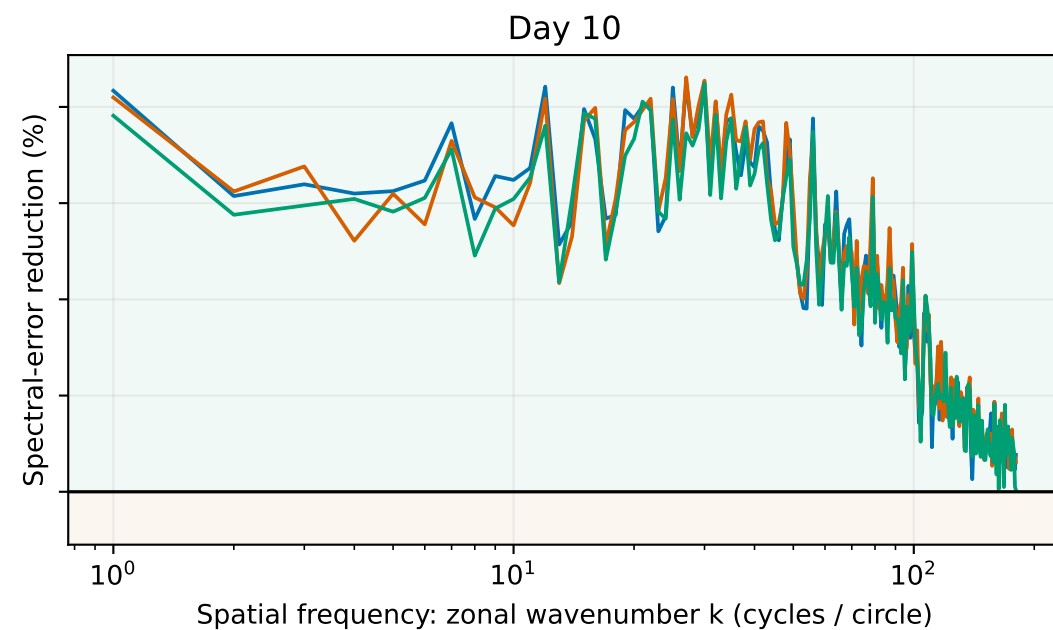
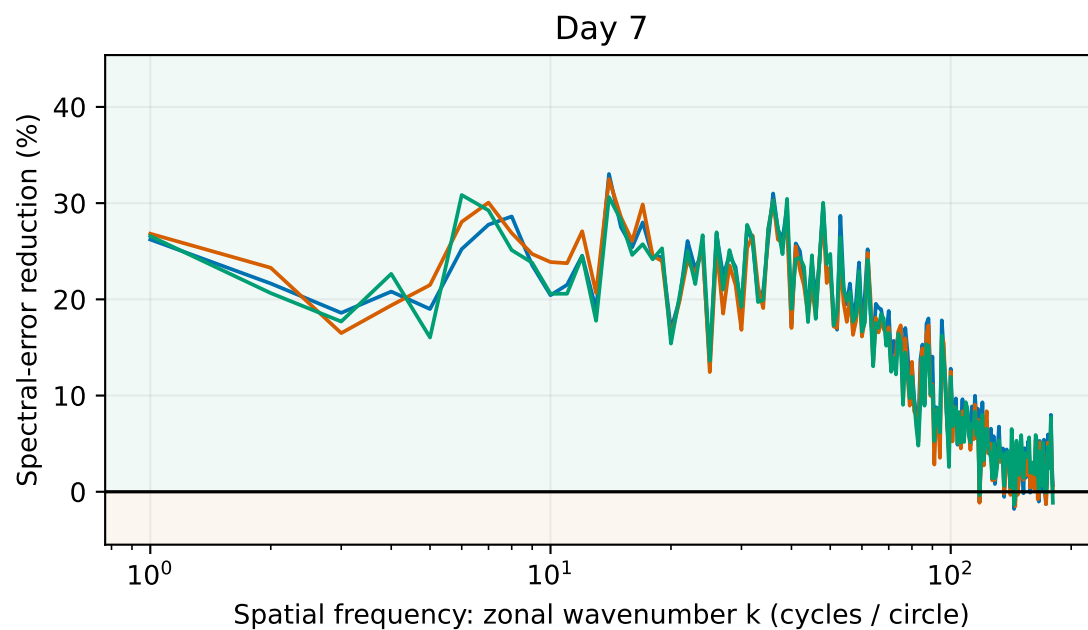
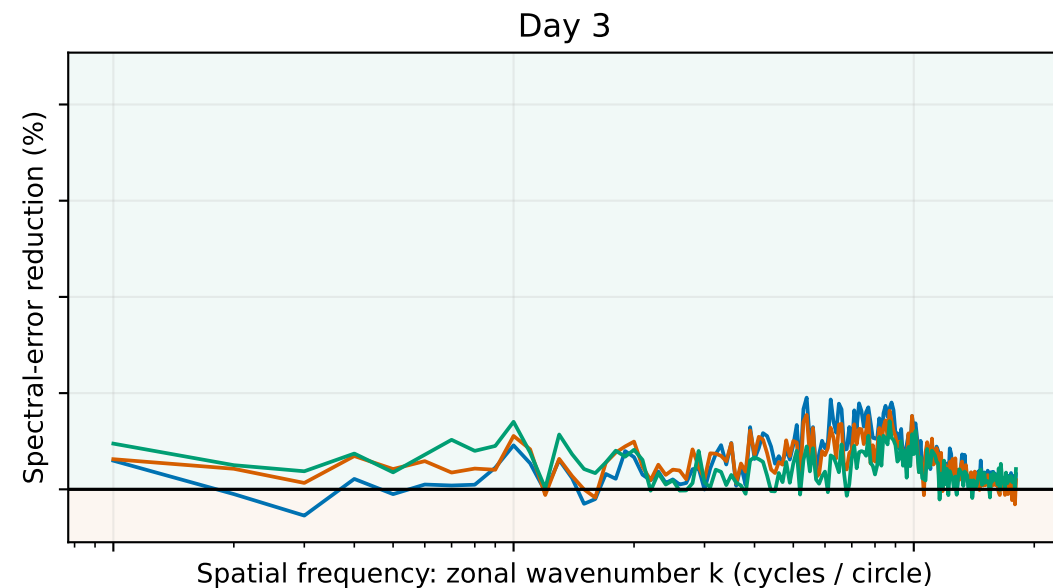
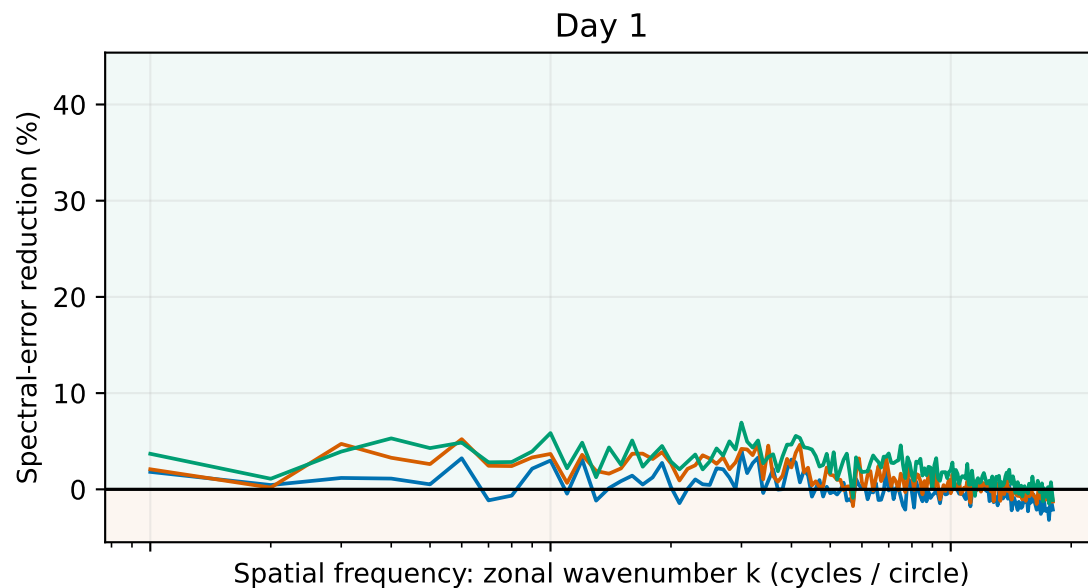
— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

TP6: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



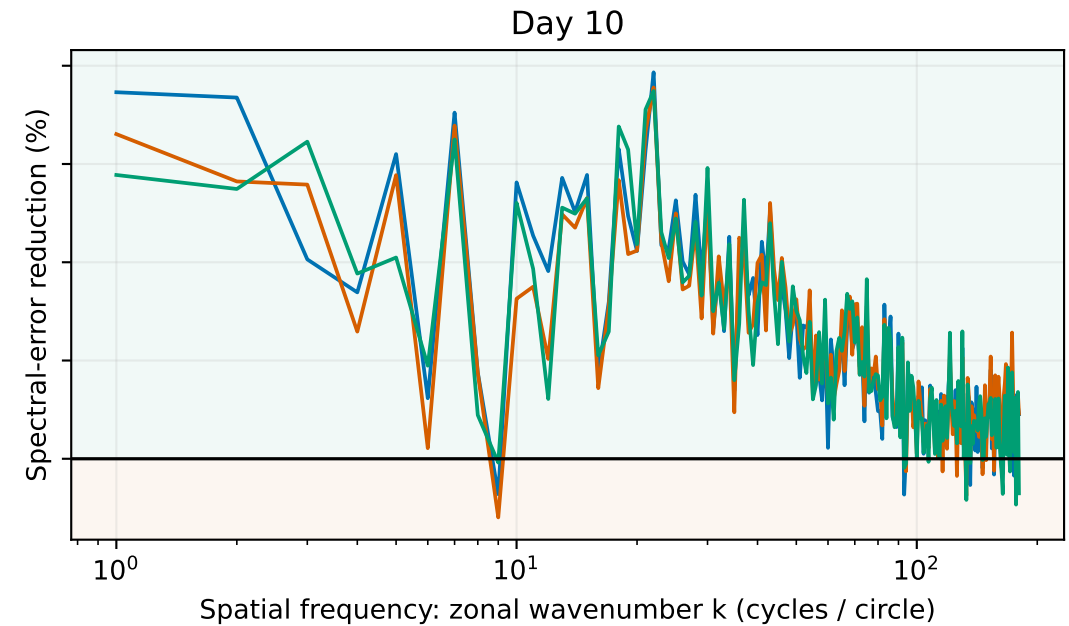
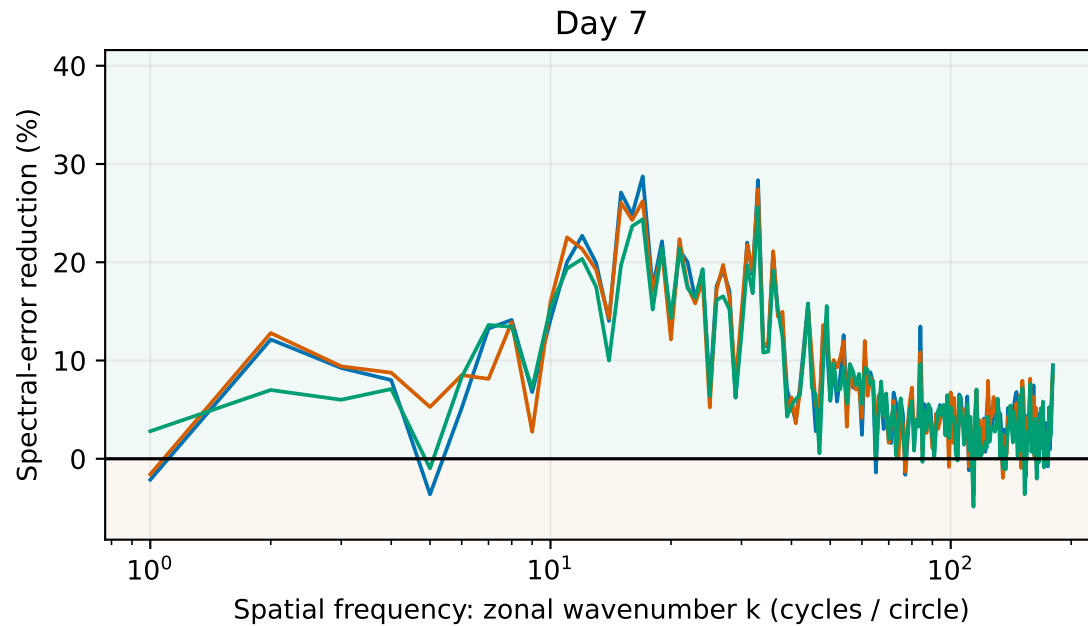
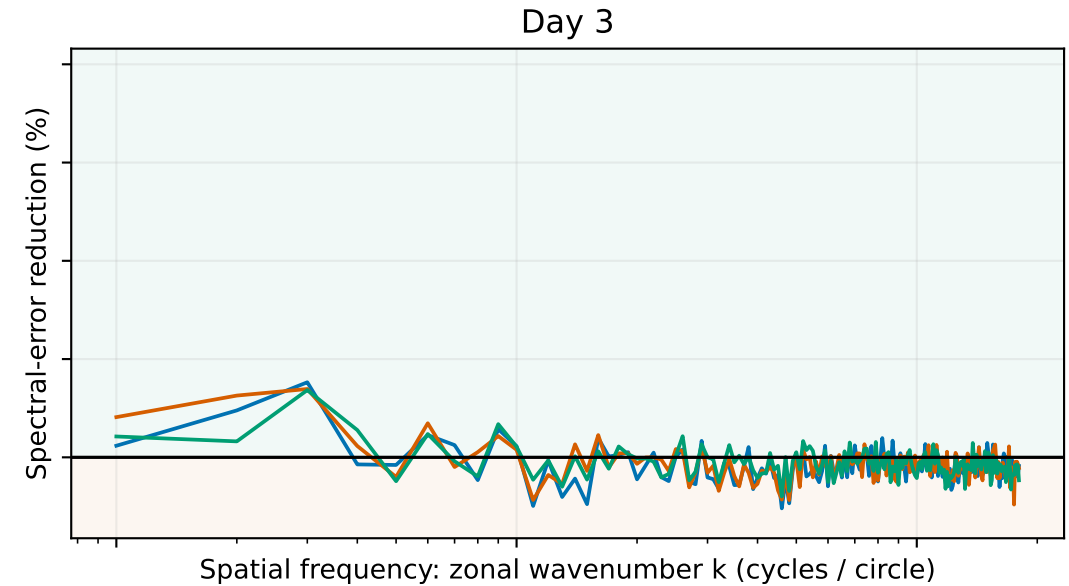
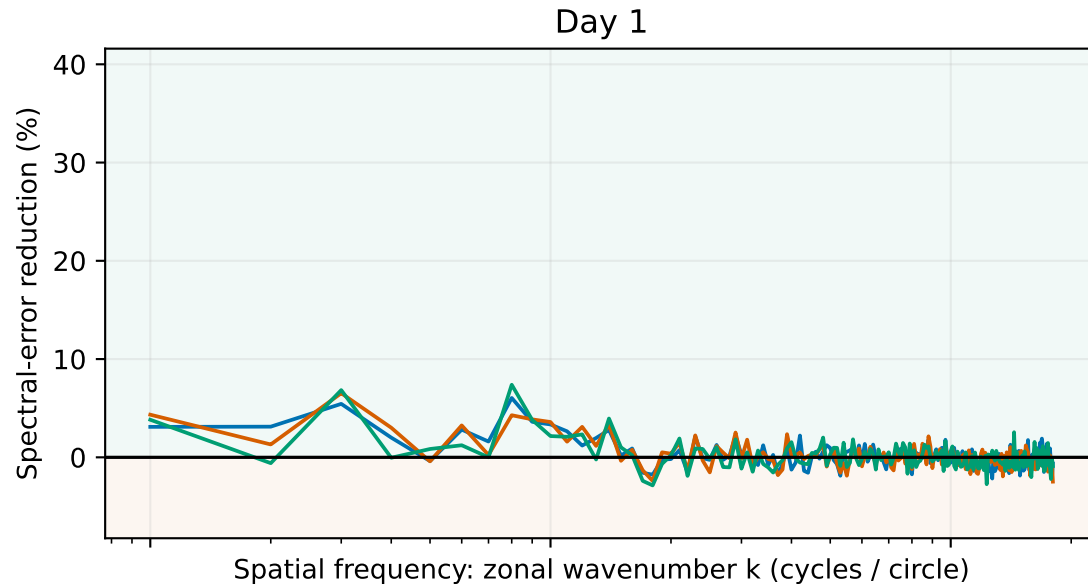
— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

T850: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



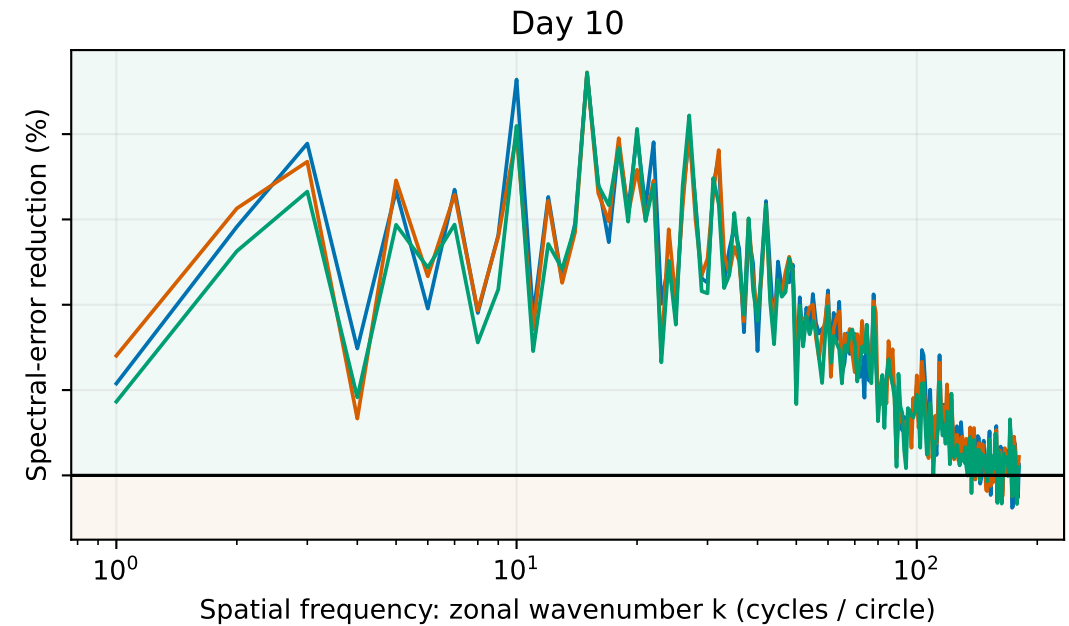
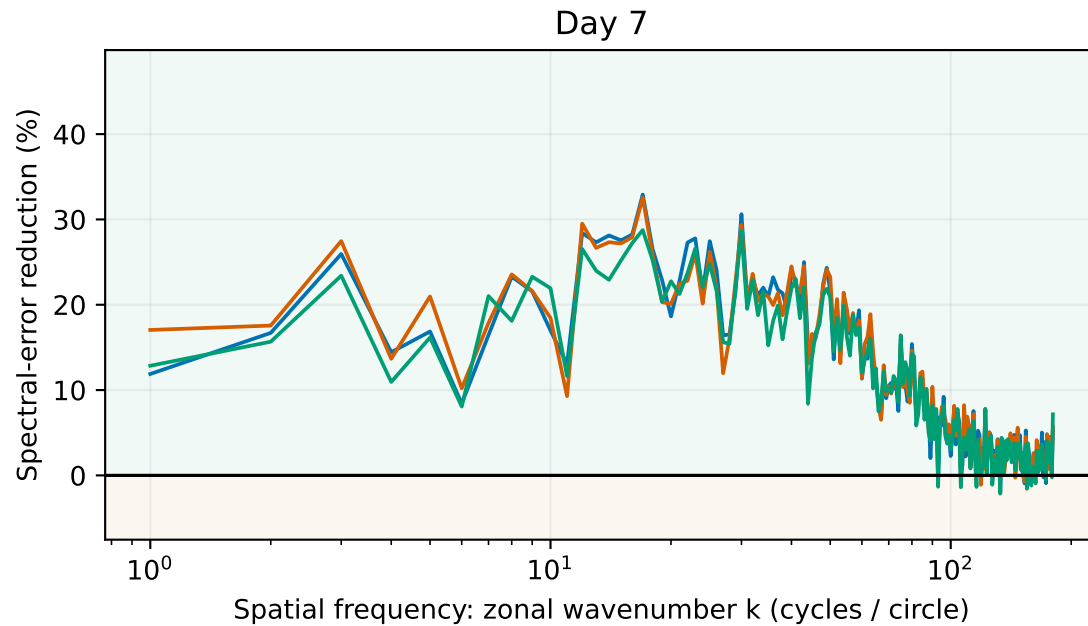
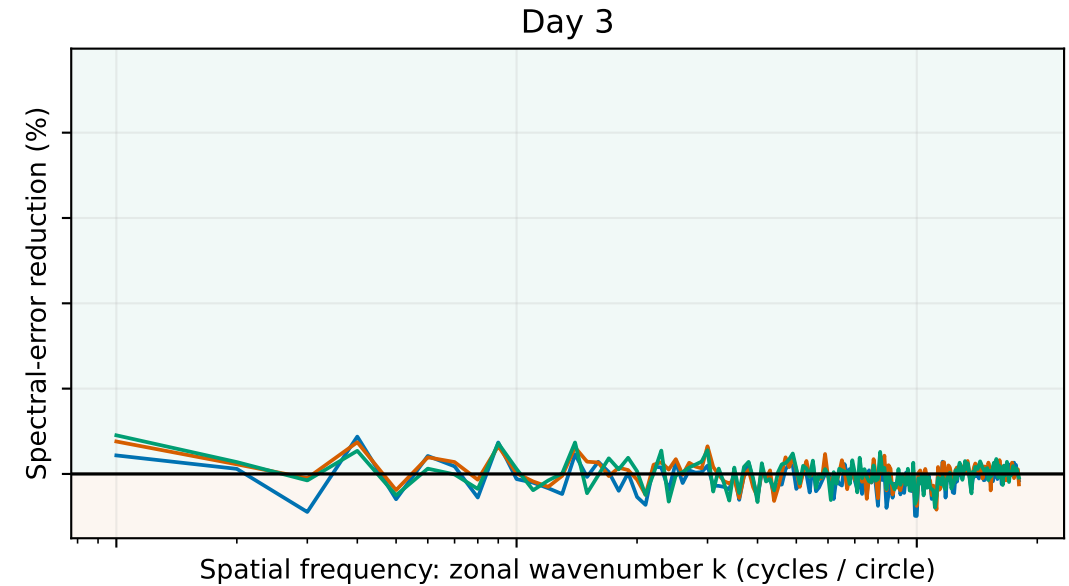
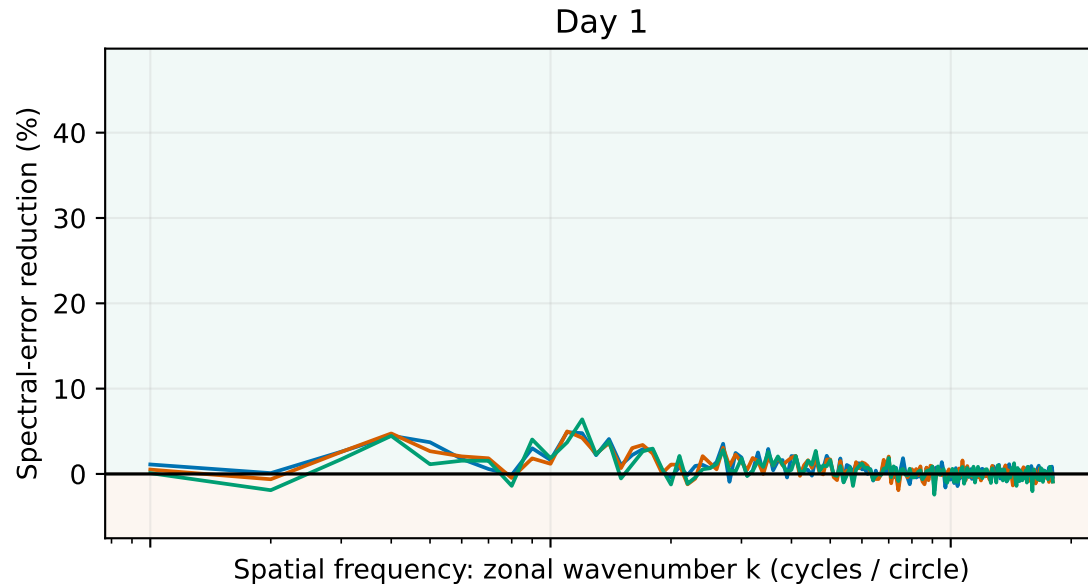
— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

Q700: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



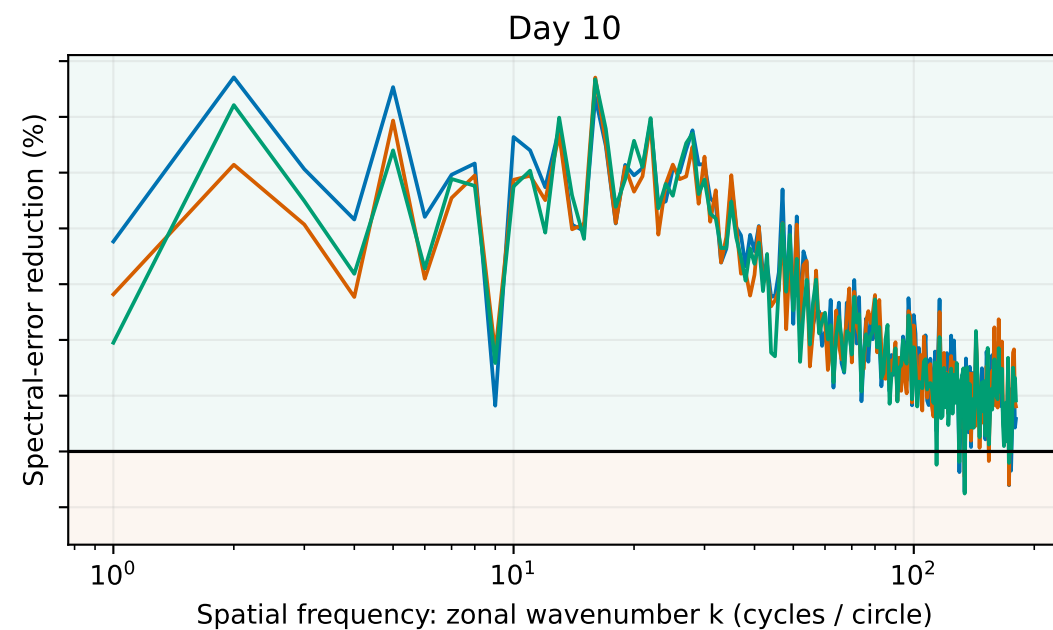
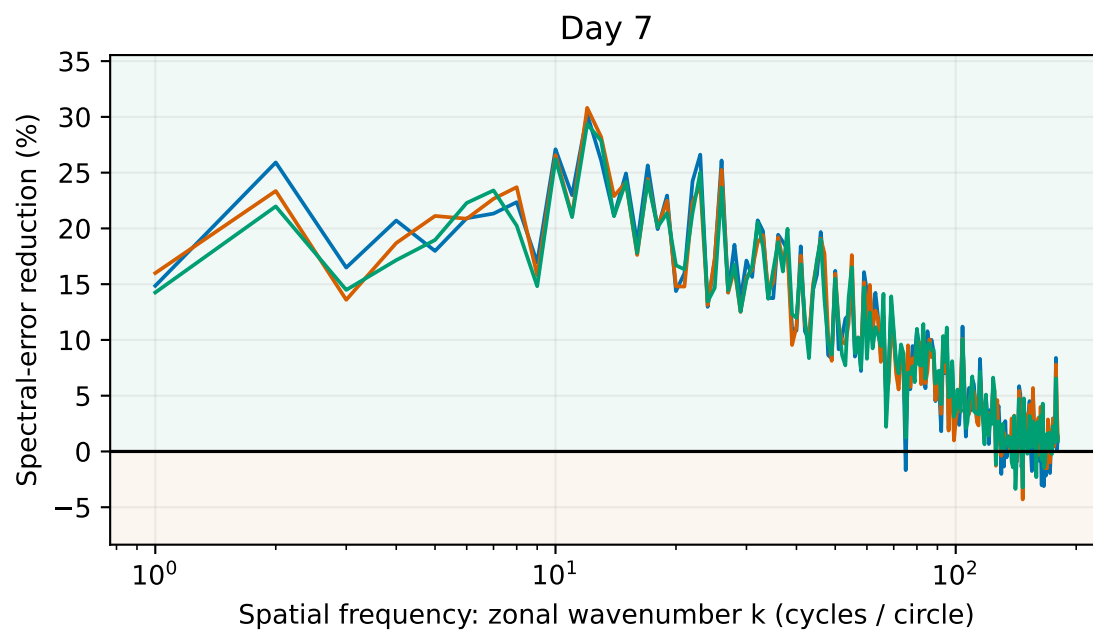
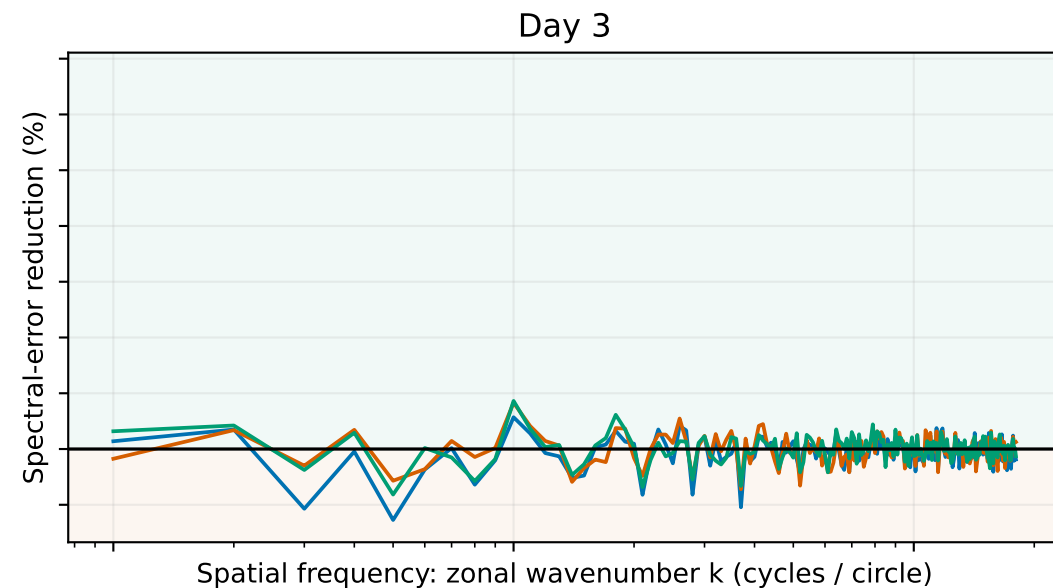
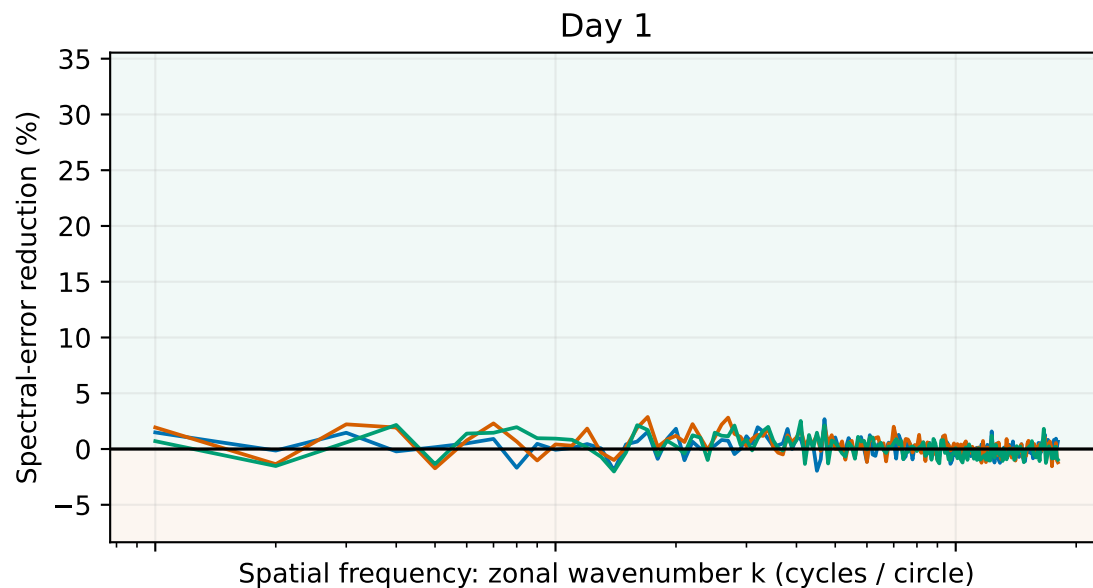
— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

U850: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



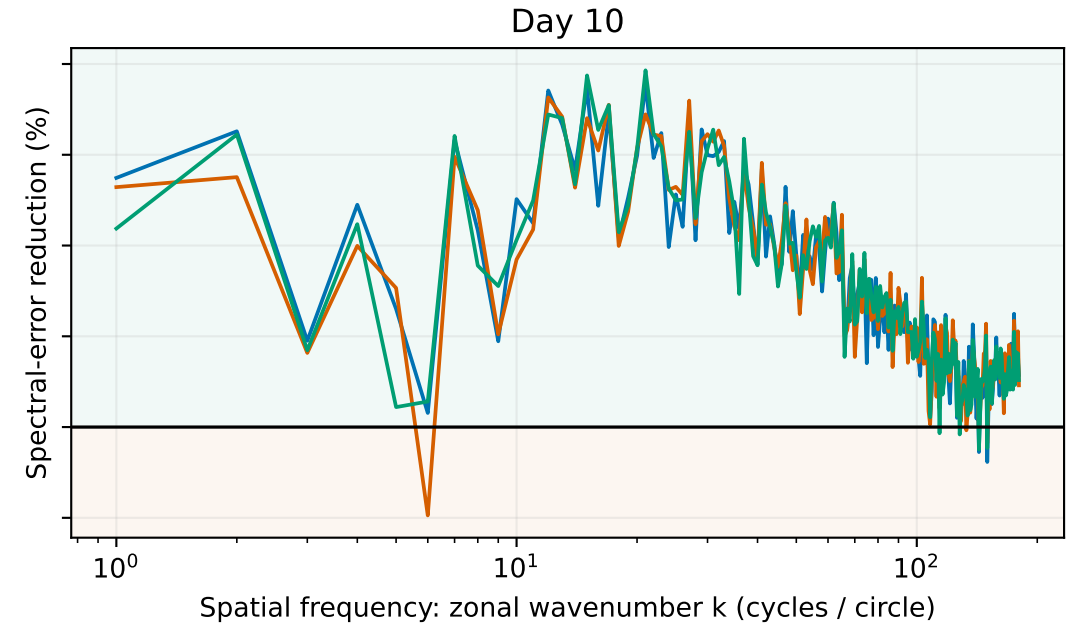
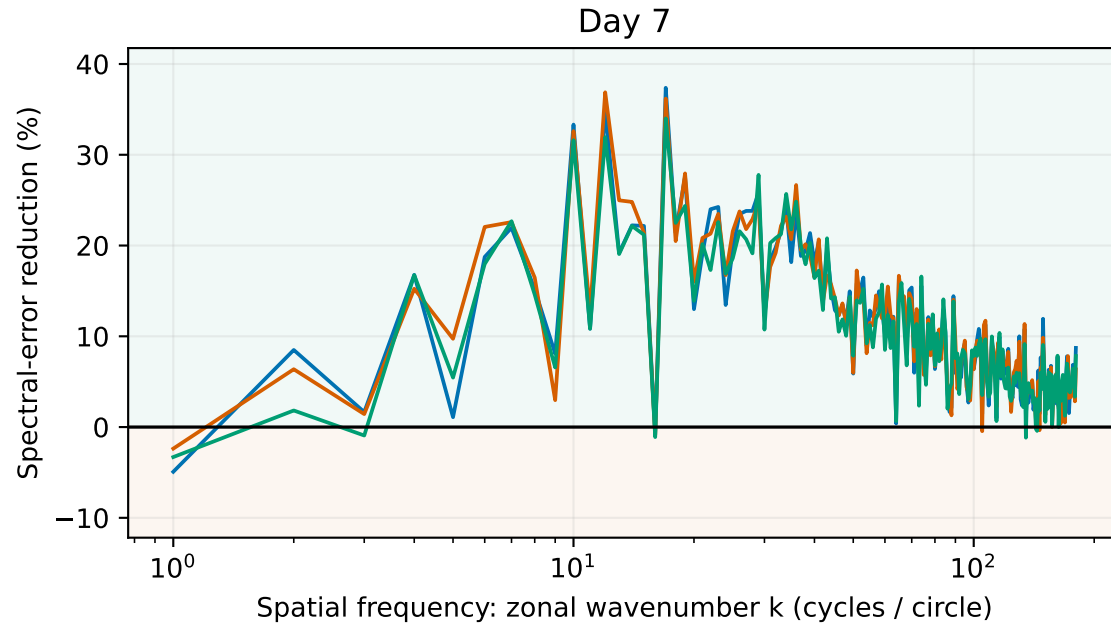
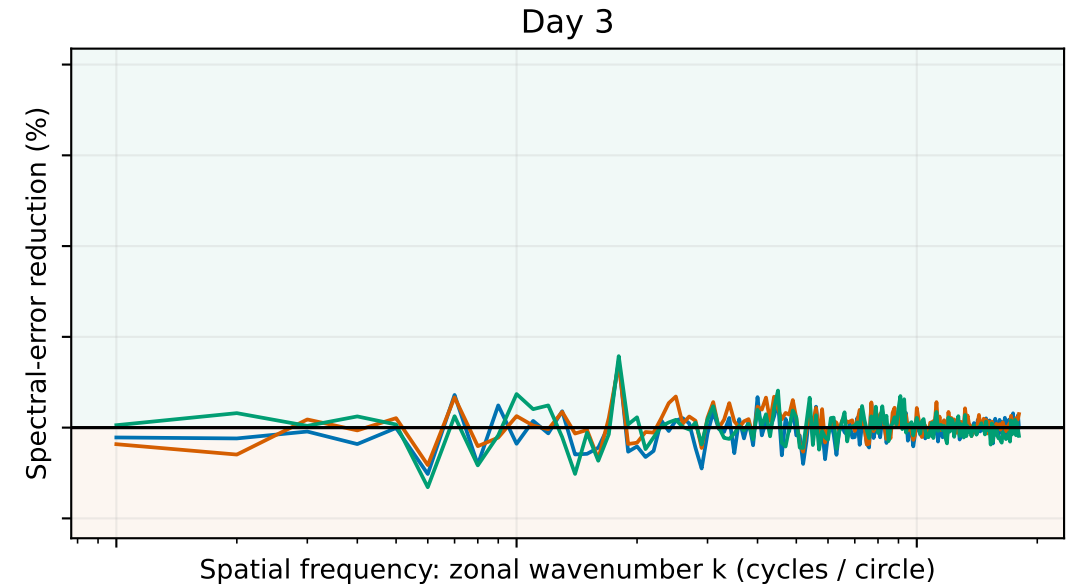
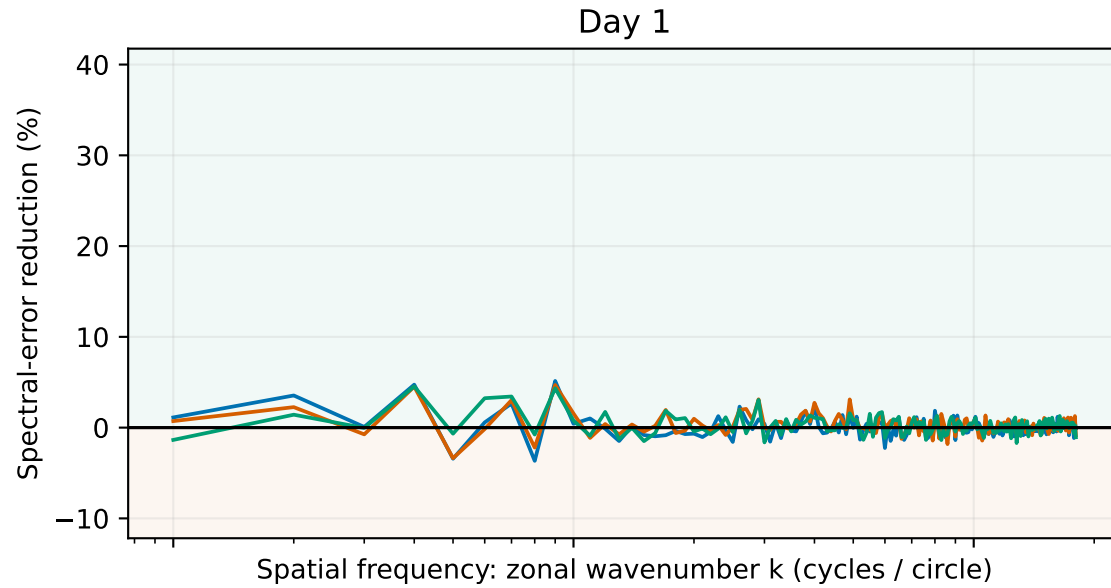
— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

V850: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



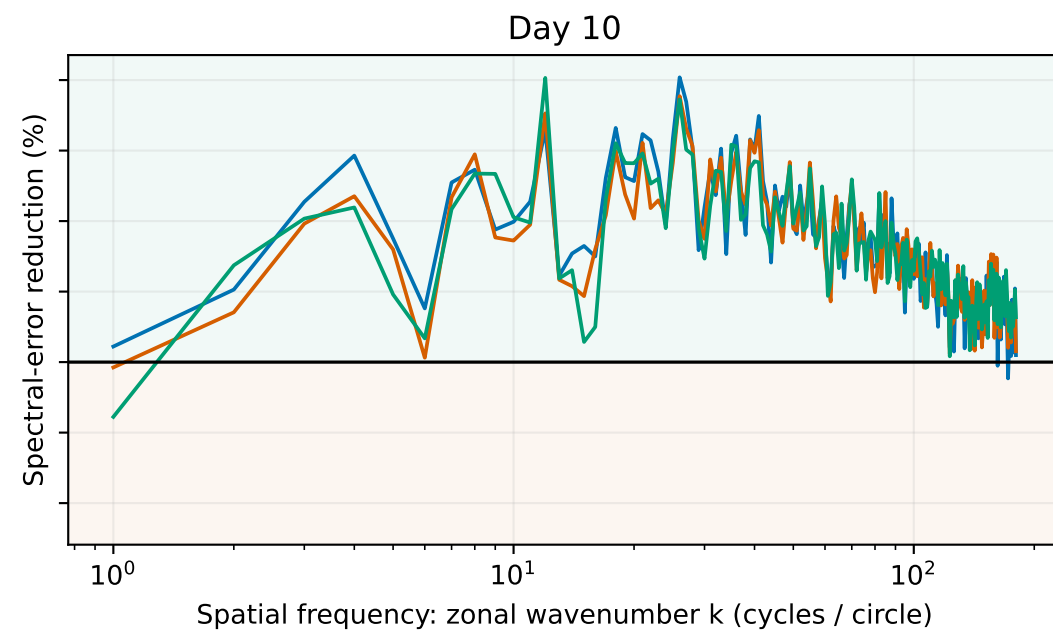
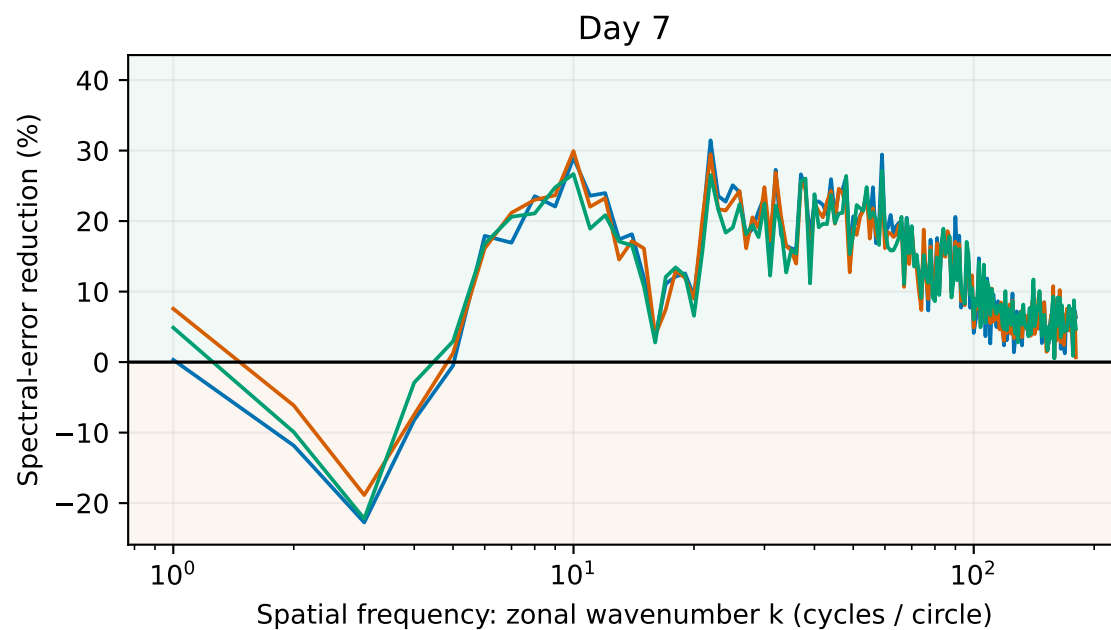
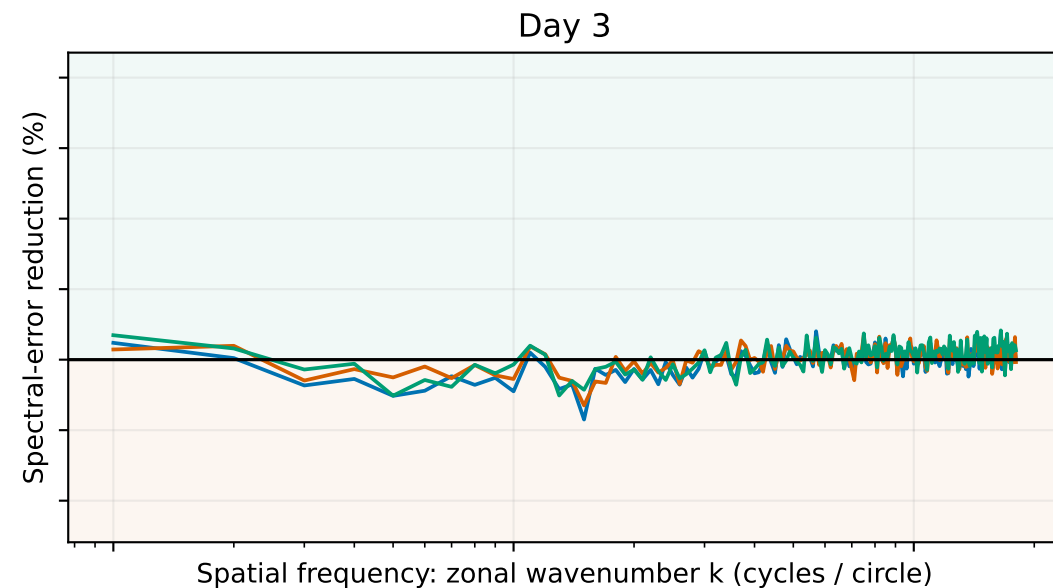
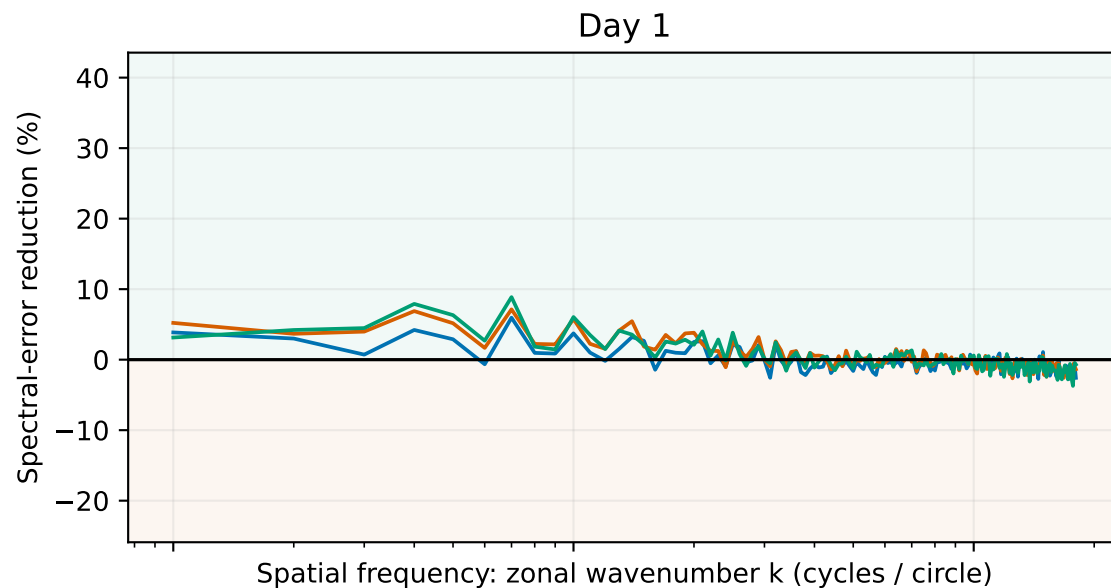
— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

WS10: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



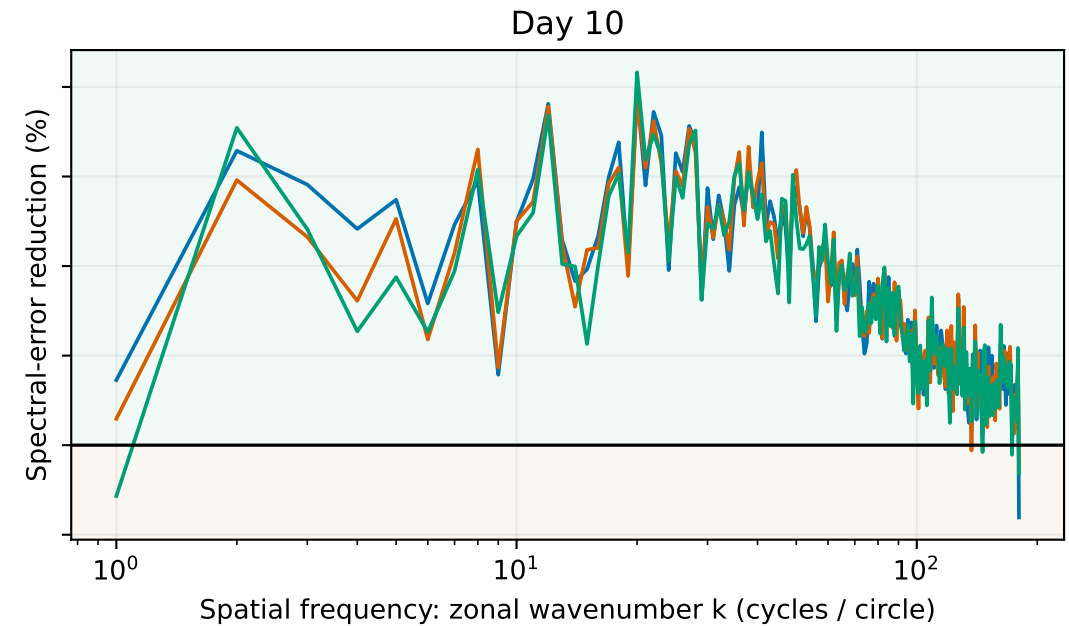
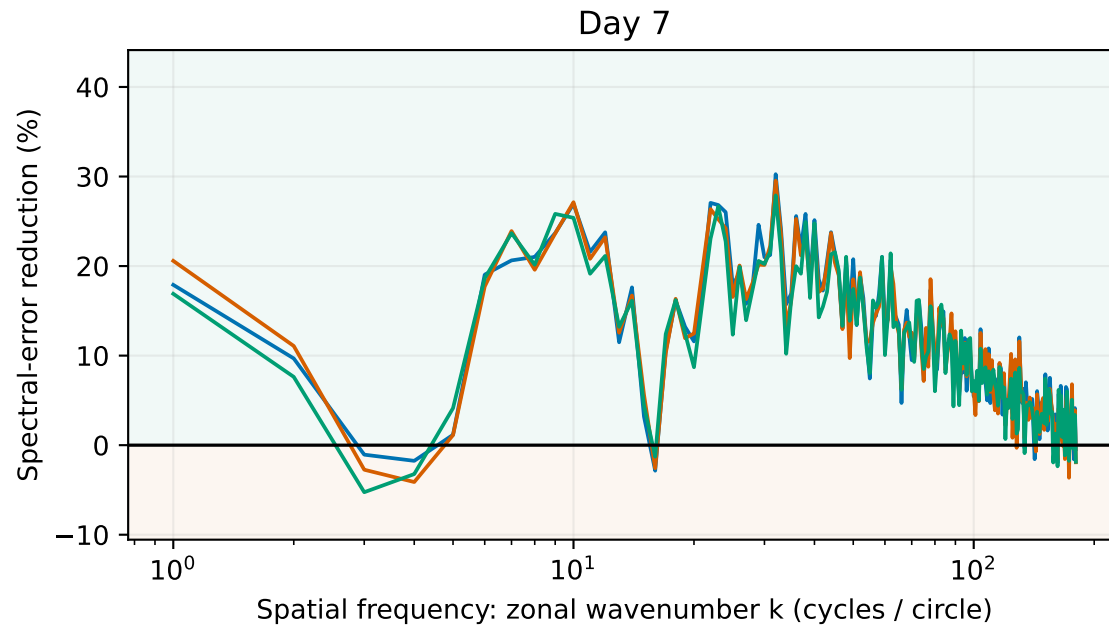
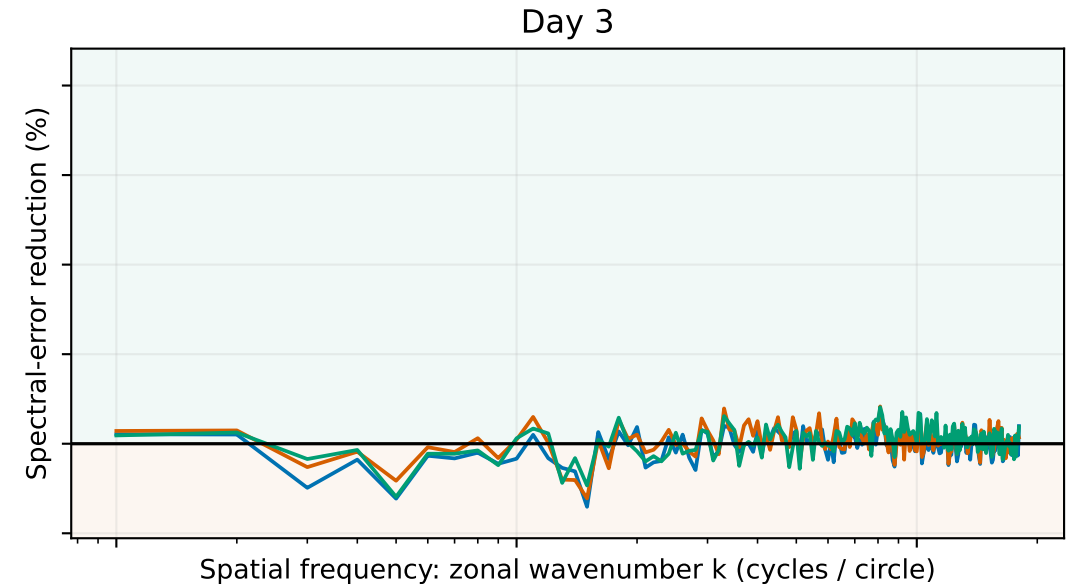
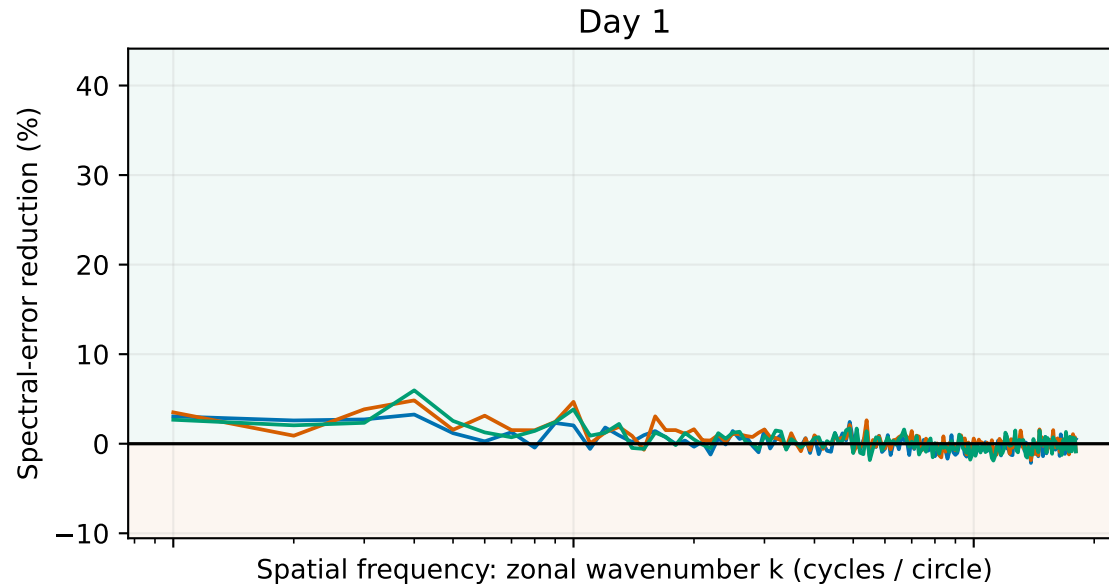
— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

WS850: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



— di16 · faster Mamba LR — di32 · joint LR — di16 · optimizer study

1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostics: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.